

Kill-only restart benchmark

When exact Poisson return cannot be implemented because the escape angle is discarded

Executive result

The Poisson kernel is still the exact mathematical answer for ordinary Brownian motion between concentric circles. This experiment asks a different engineering question: what can we do when the boundary interface reports only that a walker escaped and throws away the escape angle? In that setting, exact Poisson return is an oracle but not an eligible implementation. Centered restart remains feasible and controls the finite-boundary bias without needing the missing angle.

The information boundary: one field determines Poisson eligibility

Instrumented boundary

ESCAPED + angle theta



Poisson-conditioned return

Kill-only boundary

ESCAPED (angle discarded)



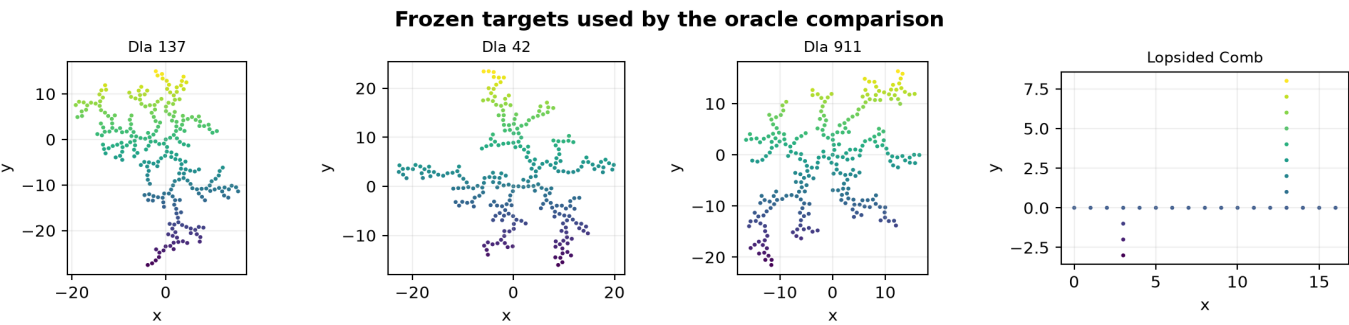
Fresh uniform or centered restart

The static evidence reuses 720 independently validated records across four asymmetric targets, nine death-radius ratios, and 20 replications. The dynamic evidence reuses the $N = 1,200$ and $N = 3,000$ four-approach runs. The exact-prefix disclosure on page 5 is intentional: those dynamic runs demonstrate calibration scheduling, not an end-to-end kill-only prefix.

1. What the interface allows

A Poisson return sample is conditioned on the point where the walker crossed the outer circle. If that angle is retained, the return correction is an $O(1)$ operation. If the angle is discarded, no later restart routine can reconstruct the same conditional law from the event flag alone. Uniform, controlled-fixed, and paper-amortized restart use fresh launches and attachment samples instead, so they remain compatible with the constrained interface.

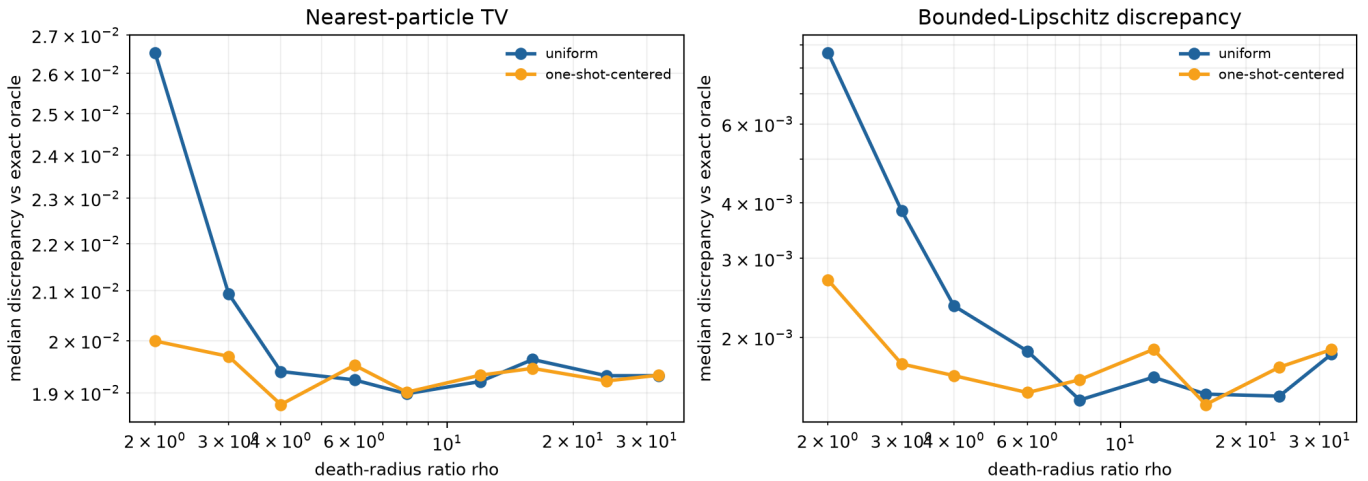
Policy	Needs escape angle	Needs attachment samples	Kill-only eligible
Exact Poisson oracle	yes	no	no
Uniform restart	no	no	yes
Per-step controlled restart	no	yes	yes
Paper-amortized control	no	yes	yes



Exact Poisson is therefore shown as the truth source and an unconstrained baseline, not as a failed competitor. The kernel remains mathematically valid on a circle; only the constrained interface prevents this component from using it.

2. Static accuracy against the oracle

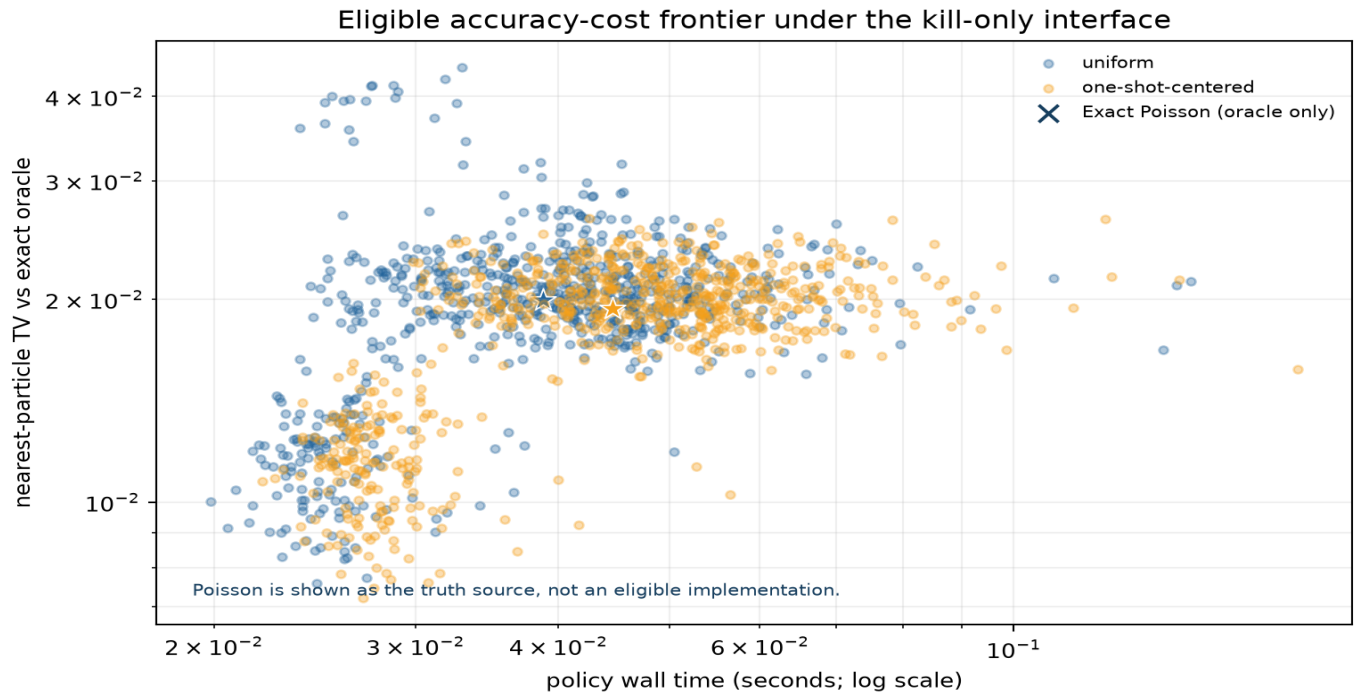
Accuracy improves as the kill boundary moves outward



The static frozen-target experiment holds the circular death boundary construction fixed and increases ρ , the death-radius to target-radius ratio. Uniform restart retains the leading directional mismatch. One-shot centering estimates the finite-boundary barycenter independently and removes that broad dipole component. The plotted values are finite partition diagnostics, not continuum total-variation proofs.

Policy	Median TV at $\rho=4$	Median TV at $\rho=16$	Interpretation
Uniform restart	0.0194	0.0196	Feasible, no calibration
One-shot centered	0.0188	0.0195	Feasible, one calibration
Exact Poisson	0	0	Oracle reference

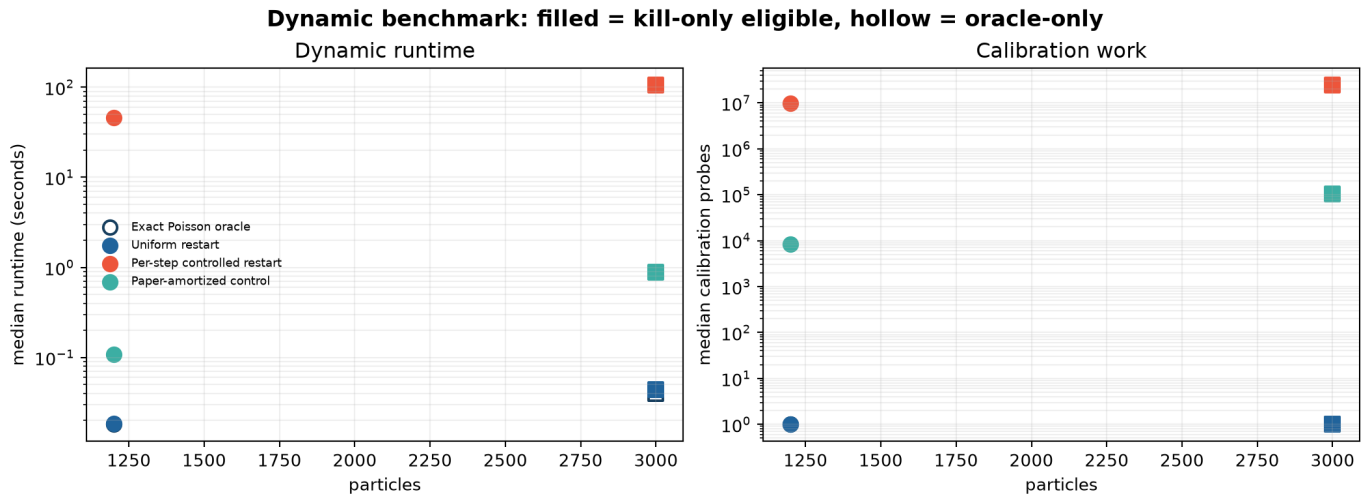
3. Eligible accuracy-cost frontier



This chart answers the practical question under the stated interface. Exact Poisson is marked oracle-only because it needs data that the kill-only adapter has destroyed. Uniform restart is the cheapest eligible policy but carries the largest finite-boundary discrepancy. Centering spends calibration work to move toward the oracle law; version 4 is designed to reuse that calibration over blocks instead of paying for a fresh search after every attachment.

A fair conclusion is therefore a feasible-frontier statement: approach 4 is valuable when the exact return information is unavailable and calibration work matters. It is not a claim that approach 4 should replace exact Poisson in a simulator that already retains escape angles.

4. Dynamic calibration work



Filled markers are kill-only eligible; hollow markers are the exact Poisson oracle. The paper-amortized schedule spreads calibration over growing blocks, while controlled-fixed repeats calibration frequently. The measured runtime and probe counts are from the existing dynamic benchmark and use an exact prefix. They are valid evidence about post-prefix scheduling cost, but they are not an end-to-end kill-only execution.

N	Policy	Median runtime (s)	Median calibration probes	Eligible
1200	Exact Poisson oracle	0.0181	0	oracle only
1200	Uniform restart	0.0186	0	yes
1200	Per-step controlled restart	45.8	9.82×10^6	yes
1200	Paper-amortized control	0.108	8.27×10^3	yes
3000	Exact Poisson oracle	0.0396	0	oracle only
3000	Uniform restart	0.0443	0	yes
3000	Per-step controlled restart	106	2.46×10^7	yes
3000	Paper-amortized control	0.883	1.02×10^5	yes

To make the complete pipeline kill-only, the exact prefix must be replaced by a separately budgeted approximate prefix. That follow-up run is a deployment validation, not silently inferred from this report.

5. Findings, limits, and reproduction

Finding: the useful scenario for approach 4 is not a circle where Poisson is somehow mathematically invalid. It is a system boundary where the escape angle is not available to the restart component. Under that contract, approach 4 provides a certified way to recover much of the lost directional information through center estimation and amortized reuse.

Limits: the paper's theorem still assumes ordinary Brownian motion and a circular death boundary. This report does not extend it to irregular boundaries, drift, heterogeneous diffusion, or arbitrary hardware. Wall times are machine-specific. The dynamic data contain an exact prefix and are labelled accordingly. The static diagnostics use finite particle and angular partitions.

Reproduction: run ``PYTHONPATH=src:. .venv/bin/python scripts/generate_kill_only_restart_report.py`` with the committed static and dynamic JSON artifacts. The generator validates the 720 static records, required policies, target count, and both dynamic summaries before writing the PDF. Render the output with Poppler for visual inspection.

Artifacts: ``scripts/generate_kill_only_restart_report.py``, ``tests/unit/test_kill_only_restart_report.py``, the generated figures under ``output/asymmetric_validation/figures_kill_only/``, and this PDF. Report date: 2026-07-26.